

TMR 4190 FINITE ELEMENT METHODS IN STRUCTURAL ANALYSIS

AUTUMN 2026

EXERCISE 3

Problem 1

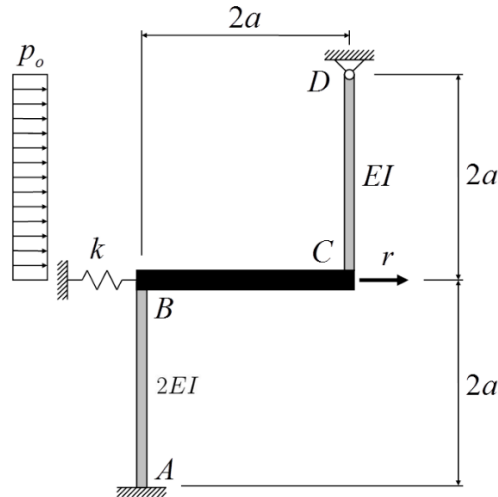


Figure 1

Figure 1 shows geometry, stiffness, loading and boundary conditions for a plane beam structure consisting of two vertical beams AB and CD connected to each other through a rigid link BC . The structure is supported by a horizontal elastic spring with stiffness $k = 5EI/8a^3$ at point B . Only bending stiffness shall be taken into account, i.e. axial and shear deformations should be neglected.

- How many are the global degrees of freedom of the configuration and how are they explicitly defined? Explain in detail the reasoning underlying your answer.
- How is the uniformly distributed load handled to be able to establish the linear stiffness relation between the external loads and the nodal displacements? Explain in detail the reasoning underlying your answer.
- What is the underlying beam theory dictating the behaviour of the given beams and why?
- If the lateral deflection w of any beam is given according to $\frac{M(x)}{EI} = \frac{d^2w}{dx^2}$, with respect their local coordinate system. Draw the analytical expression for the bending-moment distribution $M(x)$ along beam CD , assuming that the nodal degrees of freedom in the r direction are restrained and that the beam is subjected to the prescribed externally distributed load. Compare the derived nodal reactions with the corresponding data provided in the reference tables.

- e) According to your adopted method at b) and the data that have been obtained in d) establish the stiffness relation $\mathbf{K}\mathbf{r} = \mathbf{R}$, and determine $\mathbf{r} = [\mathbf{r}]$.
- f) Calculate and sketch the moment and shear force distribution for the two beams AB and CD . Comment the maxima, slope and curvature of Q and M curves for the CD beam.

Problem 2

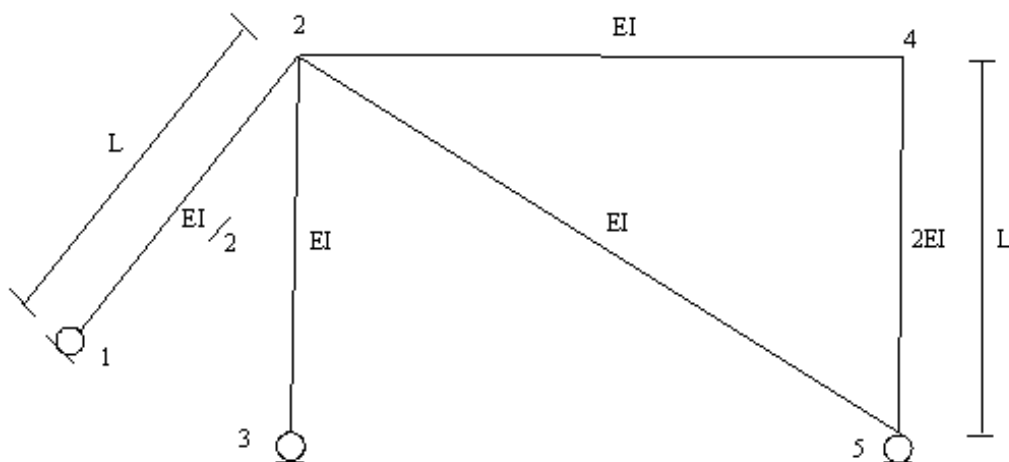


Figure 2

Figure 2 shows a plane beam structure consisting of four beam elements. Only bending stiffness shall be taken into account, i.e. axial and shear deformations should be neglected. It is given that the length of the beam 2-4 is equal to $2L$ and that the beams 2-3 and 4-5 are perpendicular to 2-4.

- a) Explain in detail what are the nodal degrees of freedoms $\mathbf{r}$ for the frame shown in Figure 2 when node 1, 3 and 5 are restrained against translations, but free to rotate. What is the difference between the nodes 1, 3, 5 and 2, 4 in your judgement?
- b) Establish the stiffness matrix $\mathbf{K}$ corresponding to $\mathbf{r}$.
- c) How is the resultant stiffness matrix $\mathbf{K}$ modified when the out-of-plane rotational degree of freedom at an arbitrary node is restrained

Problem 3

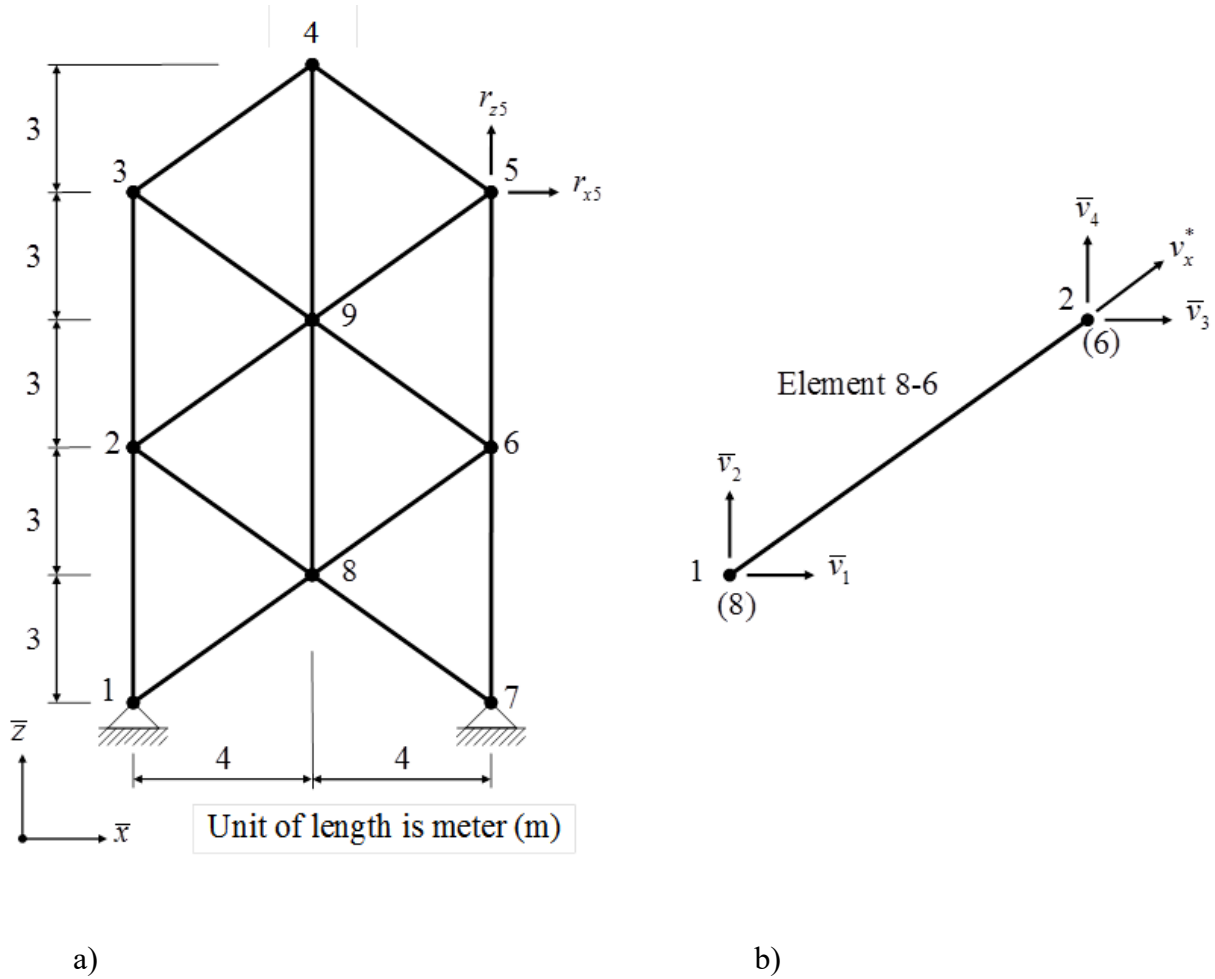


Figure 3

Figure 3a shows a plane truss structure where all elements have the same axial stiffness $EA = 50\,000$ kN. Geometry and nodal numbering is defined in Figure 3a. Each node i has two global nodal degrees of freedom (dofs) $\mathbf{r}_i = [r_{xi} \ r_{zi}]^T$, $i = 1, \dots, 9$, as shown for node 6.

- a) Establish the transformation matrix, $\mathbf{T}_{8-6}$, relating the deformational dof (i.e. the elongation of the bar) $\mathbf{v}_{8-6}^* = [v_x^*]$ to the element nodal dofs $\bar{\mathbf{v}}_{8-6} = [\bar{v}_1 \ \bar{v}_2 \ \bar{v}_3 \ \bar{v}_4]^T$ (see Figure 3b) for the bar element spanning between nodal point 8 and 6 in the truss structure in Figure 3a. The bar on top of the symbol indicates that the nodal dof $\bar{v}_i$ is referred to the global coordinates $(\bar{x}, \bar{z})$.

$$\mathbf{v}_{8-6}^* = [v_x^*] = \begin{bmatrix} T_1 & T_2 & T_3 & T_4 \end{bmatrix} \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \\ \bar{v}_4 \end{bmatrix} = \mathbf{T}_{8-6} \bar{\mathbf{v}}_{8-6}$$

- b) Establish the element stiffness matrix $\bar{\mathbf{k}}_{8-6}$, for the bar element spanning between nodal point 8 and 6, by transformation of the deformational stiffness $\mathbf{k}_{8-6}^*$.

$$\mathbf{S}_{8-6}^* = [S_x^*] = [k_{xx}^*][v_x^*] = \mathbf{k}_{8-6}^* \mathbf{v}_{8-6}^*$$

	2	3	4	5	6	8	9	
								2
								3
								4
								5
								6
								8
								9

$K_{8-6} =$